

```
!pip install opencv-python
```

```
Requirement already satisfied: opencv-python in /usr/local/lib/python3.12/dist-packages (4.13.0.92)  
Requirement already satisfied: numpy>=2 in /usr/local/lib/python3.12/dist-packages (from opencv-py
```

```
import numpy as np  
import cv2  
import matplotlib.pyplot as plt  
from tensorflow.keras.models import Sequential  
from tensorflow.keras.layers import Conv2D, MaxPooling2D, Flatten, Dense
```

```
# Create dummy images (simulating faces)  
X = np.random.rand(500, 64, 64, 3)  
  
# Gender labels (0 = Female, 1 = Male)  
gender = np.random.randint(0, 2, 500)  
  
# Age labels (random age 1-80)  
age = np.random.randint(1, 80, 500)  
  
print("Dataset Created Successfully")
```

```
Dataset Created Successfully
```

```
from tensorflow.keras.models import Model  
from tensorflow.keras.layers import Input, Conv2D, MaxPooling2D, Flatten, Dense  
  
# Input layer  
input_layer = Input(shape=(64,64,3))  
  
# CNN layers  
x = Conv2D(32, (3,3), activation='relu')(input_layer)  
x = MaxPooling2D(2,2)(x)  
  
x = Conv2D(64, (3,3), activation='relu')(x)  
x = MaxPooling2D(2,2)(x)  
  
x = Flatten()(x)  
  
# Output 1: Gender  
gender_output = Dense(1, activation='sigmoid', name='gender')(x)  
  
# Output 2: Age  
age_output = Dense(1, activation='linear', name='age')(x)  
  
# Final Model  
model = Model(inputs=input_layer, outputs=[gender_output, age_output])  
  
# Compile  
model.compile(  
    optimizer='adam',  
    loss={'gender': 'binary_crossentropy', 'age': 'mse'},  
    metrics={'gender': 'accuracy', 'age': 'mae'}  
)
```

```
history = model.fit(  
    X,  
    {'gender': gender, 'age': age},  
    epochs=5,  
    batch_size=32  
)
```

```
Epoch 1/5
16/16 ————— 4s 135ms/step - age_loss: 1014.4018 - age_mae: 26.1103 - gender_accuracy: 0.00
Epoch 2/5
16/16 ————— 2s 132ms/step - age_loss: 575.0492 - age_mae: 20.6482 - gender_accuracy: 0.00
Epoch 3/5
16/16 ————— 3s 177ms/step - age_loss: 537.8246 - age_mae: 20.1395 - gender_accuracy: 0.00
Epoch 4/5
16/16 ————— 4s 129ms/step - age_loss: 543.3546 - age_mae: 20.1151 - gender_accuracy: 0.00
Epoch 5/5
16/16 ————— 2s 127ms/step - age_loss: 527.8820 - age_mae: 20.1145 - gender_accuracy: 0.00
```

```
test_img = np.random.rand(1,64,64,3)
```

```
pred_gender, pred_age = model.predict(test_img)
```

```
print("Predicted Gender:", "Male" if pred_gender[0][0] > 0.5 else "Female")
```

```
print("Predicted Age:", int(pred_age[0][0]))
```

```
1/1 ————— 0s 85ms/step
```

```
Predicted Gender: Female
```

```
Predicted Age: 39
```